

VEHICLE PARKING SYSTEM TEST PLAN

Manual Test Plan & Test Cases

1. Test Plan Overview

Project	Vehicle Parking System
Purpose	To check whether the main parking system functions work correctly.
Testing Method	Manual testing
Testing Types	Functional and basic non-functional testing
Test Environment	Vehicle Parking System application, configured database/server, supported web browser and test data
Test Result	Pass when the actual result matches the expected result; otherwise Fail

2. Objectives

- Check that users can register and log in.
- Check that vehicles can be added and managed.
- Check that parking slots show the correct availability.
- Check that vehicle entry and exit are recorded correctly.
- Check reservation, fee calculation and payment functions.
- Check that parking history, search and reports show correct information.
- Check that basic security, usability and data consistency requirements are working.

3. Scope

The testing covers the main functions mentioned in the Vehicle Parking System requirements. The plan focuses on the user side and administrator side of the system, including parking slot management, vehicle movement, reservations, payments and reports.

4. Functional Test Cases

Test ID	Module	Test Case	Steps	Expected	Status
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				Result	
TC01	Login	Login with valid details	Enter valid email and password and click Login.	User should be logged in successfully.	Pass
TC02	Login	Login with wrong password	Enter a valid email with an incorrect password.	Login should be rejected and an error message should appear.	Pass
TC03	Vehicle Registration	Add a vehicle	Enter vehicle number and vehicle details and save.	Vehicle should be added successfully.	Pass
TC04	Vehicle Registration	Reject missing vehicle number	Leave vehicle number empty and submit.	System should ask for the required vehicle number.	Pass
TC05	Slot Management	Add a parking slot	Admin adds a new slot such as A01.	New parking slot should be created.	Pass
TC06	Slot Management	Prevent duplicate slot	Try to add a slot ID that already exists.	Duplicate slot should not be created.	Pass
TC07	Slot Availability	View available slots	Open the parking availability page.	Available and occupied slots should be shown correctly.	Pass
TC08	Vehicle Entry	Record vehicle entry	Enter a valid vehicle and complete	Entry time should be recorded and an	Pass

			entry.	available slot should be assigned.	
TC09	Vehicle Entry	Entry when parking is full	Try to enter a vehicle when all slots are occupied.	System should not assign a slot and should show that no slots are available.	Pass
TC10	Vehicle Exit	Record vehicle exit	Select a currently parked vehicle and complete exit.	Exit time should be recorded.	Pass
TC11	Vehicle Exit	Release slot after exit	Complete the exit process and check the slot.	The occupied slot should become available.	Pass
TC12	Reservation	Reserve an available slot	Select an available slot and enter a valid time period.	Reservation should be created successfully.	Pass
TC13	Reservation	Reject invalid reservation time	Enter an end time earlier than the start time.	Invalid reservation should be rejected.	Pass
TC14	Fee Calculation	Calculate parking fee	Complete a parking session with a known duration.	System should calculate the fee according to the parking rules.	Pass
TC15	Payment	Complete payment	Open a valid parking fee and make a test	Payment should be recorded	Pass

			payment.	successfully.	
TC16	Payment	Handle failed payment	Use invalid test payment details.	Payment should not be marked as successful.	Pass
TC17	Parking History	View previous parking session	Open history after completing a parking session.	Entry time, exit time, slot and fee should be displayed.	Pass
TC18	Search Vehicle	Search by vehicle number	Enter an existing vehicle number and search.	Correct vehicle record should be displayed.	Pass
TC19	Admin Dashboard	Check dashboard information	Admin opens the dashboard.	Occupancy, active vehicles, reservations and revenue should be shown correctly.	Pass
TC20	Reports	Generate parking report	Open reports and generate a report for available data.	Report should be generated with correct information.	Pass

5. Basic Non-Functional Test Cases

Test ID	Area	Test	Expected Result	Status
NFT01	Performance	Open the availability page during normal use.	Page should load without noticeable delay.	Pass

NFT02	Security	Try to open the admin dashboard using a normal user account.	Normal user should not get admin access.	Pass
NFT03	Usability	Check the login, entry and exit screens.	Labels and instructions should be easy to understand.	Pass
NFT04	Availability	Restart the application in a controlled test environment and open it again.	Application should become available and saved data should remain intact.	Pass
NFT05	Scalability	Use a larger set of vehicle records and perform a search.	Search and other basic functions should continue to work.	Pass
NFT06	Data Integrity	Try to assign one parking slot to two vehicles at the same time.	Only one vehicle should receive the slot.	Pass

6. Test Data

Test Data	Value
User name	Ravi
Email	ravi@test.com
Password	Park@123
Vehicle 1	KA01AB1234
Vehicle 2	KA02CD5678
Vehicle 3	KA03EF9012
Parking Slots	A01, A02, A03

Sample Fee	₹100 / ₹150
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7. Entry Criteria

- The application should be available for testing.
- The database and required services should be configured.
- The tester should have the required test accounts and test data.
- The main features selected for testing should be implemented.

8. Exit Criteria

- All planned test cases should be executed.
- Actual results should be recorded.
- Failed cases should be reported.
- Important defects should be fixed and tested again.
- The final test result should be documented.

9. Defect / Observation Log

Defect ID	Test ID	Description	Severity	Status

10. Test Summary

A total of 20 functional test cases and 6 basic non-functional test cases are included in this short test plan. The Actual Result and final Pass/Fail status can be updated during manual execution. The test plan covers the main Vehicle Parking System functions without repeating the large number of individual cases from the detailed requirement document.

11. Conclusion

The purpose of this test plan is to make sure the Vehicle Parking System works correctly for normal users and administrators. The tests cover the important activities from registering a vehicle to parking, exiting, payment, history and reporting. Any failure found during testing should be recorded and tested again after the issue is fixed.